

MTH166

Lecture-20

Simultaneous Differential Equations

Learning Outcomes:

To write operator form of simultaneous system of LDE and solving these.

Simultaneous Linear Differential Equations:

The system involving two first order linear differential equations in two dependent variables y_1 and y_2 and one independent variable x is called system of simultaneous linear differential equations.

Write the operator form of following system of LDE:

Problem1. $6 \frac{dy_1}{dx} + 5 \frac{dy_2}{dx} + 3y_1 + y_2 = 0, \frac{dy_2}{dx} - 5y_1 + 3y_2 = e^x$

Solution. The given system of simultaneous equations is:

$$6 \frac{dy_1}{dx} + 5 \frac{dy_2}{dx} + 3y_1 + y_2 = 0 \quad (1)$$

$$\frac{dy_2}{dx} - 5y_1 + 3y_2 = e^x \quad (2)$$

Let $D \equiv \frac{d}{dx}$, then operator form of given system can be written as:

$$(6D + 3)y_1 + (5D + 1)y_2 = 0 \quad (3) \times 5$$

$$-5y_1 + (D + 3)y_2 = e^x \quad (4) \times (6D + 3)$$

Solving these equations:

$$5(6D + 3)y_1 - 5(5D + 1)y_2 = 0$$

$$-5(6D + 3)y_1 + (6D + 3)(D + 3)y_2 = (6D + 3)e^x$$

On adding these equations, we get:

$$[(6D + 3)(D + 3) - 5(5D + 1)]y_2 = (6D + 3)e^x$$

$$\Rightarrow (6D^2 - 4D + 4)y_2 = 6e^x + 3e^x = 9e^x \quad (5)$$

To find Complimentary Function (C.F.):

$$(6D^2 - 4D + 4) = 0 \quad \Rightarrow D = \frac{1 \pm \sqrt{5}i}{3}$$

$$\Rightarrow y_c = e^{\frac{x}{3}} \left(c_1 \cos \frac{\sqrt{5}}{3} x + c_2 \sin \frac{\sqrt{5}}{3} x \right)$$

To find Particular Integral (P.I.):

$$y_p = \frac{1}{f(D)} r(x) = \frac{1}{(6D^2 - 4D + 4)} (9e^x)$$

$$\Rightarrow y_p = 9 \left[\frac{1}{(6D^2 - 4D + 4)} e^x \right]$$

$$\Rightarrow y_p = 9 \left[\frac{1}{(6(1)^2 - 4(1) + 4)} e^x \right] \quad (\text{Put } D = 1)$$

$$\Rightarrow y_p = 9 \left[\frac{1}{6} e^{2x} \right] \quad \Rightarrow y_p = \frac{3}{2} e^{2x}$$

$\therefore$ General solution is given by: $y_2 = \text{C.F.} + \text{P.I.}$

$$\text{i.e. } y_2 = y_c + y_p$$

$$\Rightarrow y_2 = e^{\frac{x}{3}} \left(c_1 \cos \frac{\sqrt{5}}{3} x + c_2 \sin \frac{\sqrt{5}}{3} x \right) + \frac{3}{2} e^{2x}$$

Again from equations (3) and (4):

$$(6D + 3)y_1 + (5D + 1)y_2 = 0 \quad (3) \times (D + 3)$$

$$-5y_1 + (D + 3)y_2 = e^x \quad (4) \times (5D + 1)$$

Solving these equations:

$$(6D + 3)(D + 3)y_1 + (5D + 1)(D + 3)y_2 = 0$$

$$-5(5D + 1)y_1 + (5D + 1)(D + 3)y_2 = (5D + 1)e^x$$

Subtracting these equations, we get:

$$[(6D + 3)(D + 3) + 5(5D + 1)]y_1 = (5D + 1)e^x$$

$$\Rightarrow (6D^2 + 46D + 14)y_1 = 6 e^x$$

Try it yourself.

Polling Question:

The operator form of $3\frac{dy_1}{dx} + 2y_1 + y_2 = e^{-x}$, $\frac{dy_1}{dx} + \frac{dy_2}{dx} - 2y_1 + 3y_2 = x$ is:

(A) $(3D + 2)y_1 + y_2 = 0$, $(D - 2)y_1 + (D + 3)y_2 = 0$

(B) $(3D + 2)y_1 + y_2 = e^{-x}$, $(D - 2)y_2 + (D + 3)y_1 = x$

(C) $(3D + 2)y_1 + y_2 = e^{-x}$, $(D - 2)y_1 + (D + 3)y_2 = x$

Problem2. $3 \frac{dy_1}{dx} + 2y_1 + y_2 = e^{-x}, \frac{dy_1}{dx} + \frac{dy_2}{dx} - 2y_1 + 3y_2 = x$

Solution. The given system of simultaneous equations is:

$$3 \frac{dy_1}{dx} + 2y_1 + y_2 = e^{-x} \quad (1)$$

$$\frac{dy_1}{dx} + \frac{dy_2}{dx} - 2y_1 + 3y_2 = x \quad (2)$$

Let $D \equiv \frac{d}{dx}$, then operator form of given system can be written as:

$$(3D + 2)y_1 + y_2 = e^{-x} \quad (3) \times (D + 3)$$

$$(D - 2)y_1 + (D + 3)y_2 = x \quad (4) \times 1$$

Solving these equations:

$$(3D + 2)(D + 3)y_1 + (D + 3)y_2 = (D + 3)e^{-x}$$

$$(D - 2)y_1 + (D + 3)y_2 = x$$

Subtracting these equations, we get:

$$[(3D + 2)(D + 3) - (D - 2)]y_1 = (D + 3)e^{-x} - x$$

$$\Rightarrow (3D^2 + 10D + 8)y_1 = 2e^{-x} - x \quad (5)$$

To find Complimentary Function (C.F.):

$$(3D^2 + 10D + 8) = 0 \quad \Rightarrow D = -2, -\frac{4}{3}$$

$$\Rightarrow y_c = \left(c_1 e^{-2x} + c_2 e^{-\frac{4}{3}x} \right)$$

To find Particular Integral (P.I.):

$$y_p = \frac{1}{f(D)} r(x) = \frac{1}{(3D^2 + 10D + 8)} (2e^{-x} - x)$$

$$\Rightarrow y_p = 2 \left[\frac{1}{(3D^2 + 10D + 8)} e^{-x} \right] - \left[\frac{1}{(3D^2 + 10D + 8)} x \right]$$

$$\Rightarrow y_p = 2 \left[\frac{1}{(3(-1)^2 + 10(-1) + 8)} e^{-x} \right] - \left[\frac{1}{8 \left(1 + \left(\frac{10D + 3D^2}{8} \right) \right)} x \right]$$

$$\Rightarrow y_p = 2 \left[\frac{1}{1} e^{-x} \right] - \frac{1}{8} \left[\left(1 + \left(\frac{10D + 3D^2}{8} \right) \right)^{-1} x \right]$$

$$\Rightarrow y_p = 2e^{-x} - \frac{1}{8} \left(x - \frac{10}{8} \right)$$

∴ General solution is given by: $y_1 = \text{C.F.} + \text{P.I.}$

i.e. $y_1 = y_c + y_p$

$$\Rightarrow y_1 = \left(c_1 e^{-2x} + c_2 e^{-\frac{4}{3}x} \right) + 2e^{-x} + \frac{1}{8} \left(x - \frac{10}{8} \right)$$

Again from equations (3) and (4):

$$(3D + 2)y_1 + y_2 = e^{-x} \quad (3) \times (D - 2)$$

$$(D - 2)y_1 + (D + 3)y_2 = x \quad (4) \times (3D + 2)$$

Solving these equations:

$$(3D + 2)(D - 2)y_1 + (D - 2)y_2 = (D - 2)e^{-x}$$

$$(3D + 2)(D - 2)y_1 + (3D + 2)(D + 3)y_2 = (3D + 2)x$$

Subtracting these equations, we get:

$$\begin{aligned}
& [(3D + 2)(D + 3) - (D - 2)]y_2 = (3D + 2)x - (D - 2)e^{-x} \\
& \Rightarrow (3D^2 + 10D + 8)y_2 = (3 + 2x) + 3e^{-x} \qquad (6)
\end{aligned}$$

To find Complimentary Function (C.F.):

$$(3D^2 + 10D + 8) = 0 \qquad \Rightarrow D = -2, -\frac{4}{3}$$

$$\Rightarrow y_c = \left(c_3 e^{-2x} + c_4 e^{-\frac{4}{3}x} \right)$$

To find Particular Integral (P.I.):

$$y_p = \frac{1}{f(D)} r(x) = \frac{1}{(3D^2+10D+8)} (3e^{-x} + (2x + 3))$$

$$\Rightarrow y_p = 3 \left[\frac{1}{(3D^2+10D+8)} e^{-x} \right] + \left[\frac{1}{(3D^2+10D+8)} (2x + 3) \right]$$

$$\Rightarrow y_p = 3 \left[\frac{1}{(3(-1)^2+10(-1)+8)} e^{-x} \right] + \left[\frac{1}{8 \left(1 + \left(\frac{10D+3D^2}{8} \right) \right)} (2x + 3) \right]$$

$$\Rightarrow y_p = 3 \left[\frac{1}{1} e^{-x} \right] + \frac{1}{8} \left[\left(1 + \left(\frac{10D+3D^2}{8} \right) \right)^{-1} (2x + 3) \right]$$

$$\Rightarrow y_p = 3e^{-x} + \frac{1}{8} \left((2x + 3) - \frac{10}{4} \right)$$

∴ General solution is given by: $y_2 = \text{C.F.} + \text{P.I.}$

i.e. $y_2 = y_c + y_p$

$$\Rightarrow y_2 = \left(c_3 e^{-2x} + c_4 e^{-\frac{4}{3}x} \right) + 2e^{-x} + \frac{1}{8} \left((2x + 3) - \frac{10}{4} \right)$$

